

The empirical recurrence for OEIS A202801, proved by transfer matrix

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Abstract

OEIS A202801 counts arrays of width 7 over $\{0, 1\}$ in which every cell of a named value has a neighbour of that same value. The entry carries an empirical recurrence of order 96 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: what a cell asks reaches one row above it and one row below, so the condition on a whole row is settled by 3 consecutive rows and the count is a walk count on $S = 16513$ states.

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1 The conjecture

OEIS A202801 is “Number of $n \times 7$ binary arrays with every one adjacent to another one horizontally or vertically.”. It has offset 1 and begins

37, 5854, 581518, 57127796, 5730640104, 573985849362, 57451983566596, 5751152791799586, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 104*a(n-1) - 1509*a(n-2) + 112845*a(n-3) - 549886*a(n-4) + 45197781*a(n-5) + 954444961*a(n-6) + 6012871096*a(n-7) + 72065190780*a(n-8) - 403254005060*a(n-9) + 9378286042322*a(n-10) - 135166611568332*a(n-11) - 4746618772655462*a(n-12) + 16750296332970178*a(n-13) + 641491247046472171*a(n-14) - 581930862785744235*a(n-15) - 49991110179300244938*a(n-16) - 29003429867913627250*a(n-17) + 2663525173868076038093*a(n-18) + 4406874281358922155730*a(n-19) - 104106044932848817474919*a(n-20) - 26586766049617261206547021) + 3086402955290440664148659*a(n-22) + 10540986401841044742983153*a(n-23) - 70218232510533503479549216*a(n-24) - 307245955700207265201868114*a(n-25) + 1212383133878529390626) + 6866570110100841851876314510*a(n-27) - 15012333083873389551099016781*a(n-28) - 118647402020016868727278957170*a(n-29) + 106189463033142182852842165411*a(n-30) + 1540897216831050214590543591279*a(n-31) + 262440887754236012862617053019*a(n-32) - 13518096646340292601685166108359*a(n-33) - 16546338300471680962140597351151*a(n-34) + 47765424006768451986599376467562*a(n-35) + 163806463080913659915154514986896*a(n-36) + 536742812326340611899510970845584*a(n-37) + 174526235539143097611558927527224*a(n-38) - 7192121633047695793784665688583428*a(n-39) - 16569774194220947979434206188542512*a(n-40) - 9658267954384872764093864664281664*a(n-41) + 30492528686215262217595198610977008*a(n-42) + 504791992429513047738159990183241728*a(n-43) + 1419342473741519645007077832134996928*a(n-44) + 1810893328523449947797125818346819840*a(n-45) + 2939088926460403031996171321206102784*a(n-46) - 20255724084329967352729754992289424384*a(n-47) - 76166005674682479831612723739141989376*a(n-48) - 248974601455563952176848405805301055488*a(n-49) - 833285621196686767464861469941415133184*a(n-50) - 1432216467613919967438181586797726040064*a(n-51) - 483808191675027909179772168599186079744*a(n-52) - 5659868179140363210847357530923549261824*a(n-53) - 200235194821873528231063381557352050196$

54) -17515257336613623462898975747336808169472*a(n-55) -65823364170541006525696256612951623991
 56) -46182151861365787992372020009592554520576*a(n-57) -18203652753276318731331788935835372067
 58) -110029209391952592118575457814138150977536*a(n-59) -4364315609163323168245900599242427497
 60) -241511299748660190480314157721498329022464*a(n-61) -9175050733460492484154947659440047280
 62) -475975119549559417737600877672639894650880*a(n-63) -1693953068177868014953753685104003909
 64) -79697085049659230997409853074104048733388*a(n-65) -2745442504517809978504228087185186750
 66) -1046183648475320719686894778767580143288320*a(n-67) -390902568087222546937120293886955338
 68) -898371131098940347182035562985602666201088*a(n-69) -4877748294732680516617956367536404276
 70) -43824310362633631315277670432123964620800*a(n-71) -52398967874367955760385277273014195579
 72) +1439252103318783464525519935652231001407488*a(n-73) -45910121207399989184976083883656090
 74) +2847398570181152993261160773330306003369984*a(n-75) -28322385168848166052408109176396642
 76) +3174244463978844415817485725828191526846464*a(n-77) -54353066149775978355818467981151631
 78) +2000756614600189176010604576772800565477376*a(n-79) +1136274369703102707634236269797065
 80) +155595201809921575561018733891128588763136*a(n-81) +1432418898106254927833056855924443
 82) -903502390892343797605592922760949031501824*a(n-83) +712152927879935508741944815907733589
 84) -647766834040807798710347906921632590987264*a(n-85) -1058094309399317212572949771340008652
 86) +25589869503210297642199620181684268302336*a(n-87) -1924684453334405656876451709288427951
 88) +198411972776975000023937066469840421650432*a(n-89) -624918549306085365179281838327693699
 90) +32246263876992604929208474909958222118912*a(n-91) +116234906753695334168578011077690534
 92) -24710732671609117654526284507339967954944*a(n-93) +3398835985222029887015034653136841080
 94) +2945569238659373574354836445581243580416*a(n-95) -68056473384187692692674921486353642291
 96)

As of the “Last modified” line on the live entry (Oct 03 2025 09:55:52, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The arrays have 7 columns and a growing number of rows, with entries in $\{0, 1\}$. Fix the offsets the entry names,

$$\Delta = \{(-1, 0), (0, -1), (0, 1), (1, 0)\},$$

and for a cell (i, j) write

$$S(i, j) = \{(i + \delta_1, j + \delta_2) : (\delta_1, \delta_2) \in \Delta\} \cap (\text{the cells of the array})$$

for the neighbours it actually has. Only offsets landing inside the array count, so a cell on the border simply has fewer neighbours.

The condition is that every cell equal to 1 has a neighbour equal to 1:

$$x(i, j) = 1 \implies \exists (a, b) \in S(i, j) \ x(a, b) = 1.$$

Cells of any other value are under no condition.

3 A window of 3 rows

The offsets reach one row up and one row down, so everything the condition asks of one row is decided by that row together with one row above it and one row below — a window of 3 consecutive rows, and nothing wider. Take the array one row at a time and let the state be the last 2 of them, with a distinguished start standing for the array before its first row, so that a walk of n steps is an array of n rows.

Appending a row completes exactly one window, and the row it settles is the one just above the bottom of that window; a step is allowed when that row passes. Each row of the finished array is therefore tested exactly once.

When the array stops, the last row of it has never been tested, because the conditions reach one row below. So the end vector is NOT all ones: a state is accepted exactly when those rows pass with nothing below them, which is decided by running the same test on the window with absent rows appended — the same rule that governs a cell at the bottom edge in Section 2. Thus

$$a(n) = \iota^\top M^j \tau, \quad S = 16513,$$

for the transfer matrix M on those S states. The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C -finite.

4 The proof

Lemma 1. *Let $q(t) = t^{96} - \sum_i c_i t^{96-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+96} - \sum_i c_i M^{j+96-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) = 104a(n-1) - 1509a(n-2) + 112845a(n-3) - 549886a(n-4) + 45197781a(n-5) + 95444961a(n-6) + 6012$$

for every $n > 95$.

Proof. The vector $w = q(M)\tau$ was formed by 96 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 95 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: every array of width 7 over $\{0, 1\}$ with a few rows was written out cell by cell and each cell tested against the neighbours it has. No window, no state and no end vector are involved in that computation, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A202801.

- [2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.